## ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 0 | She returns to the starting point. |
| 2 | **(d)** acceleration | Change in velocity per time. |
| 3 | **(b)** velocity, acceleration and displacement | Third equation. |
| 4 | **(c)** greater than or equal to the magnitude of displacement | Path length is at least the straight-line gap. |
| 5 | **(d)** displacement | Product of v and t. |
| 6 | **(d)** 10 m/s squared | g about 9.8. |
| 7 | **(b)** vector quantity | It has magnitude and direction. |
| 8 | **(d)** at rest | Distance does not change. |
| 9 | **(b)** 20 m/s | v = gt. |
| 10 | **(c)** scalar quantity | Only magnitude. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Distance: total path length, scalar (1); Displacement: shortest straight-line gap with direction, vector (1). | 1+1 |
| 12 | Rate of change of velocity (1); m/s squared (1). | 1+1 |
| 13 | Direction of velocity changes continuously (1); Change in velocity means acceleration (1). | 1+1 |
| 14 | Speed: distance/time, scalar (1); Velocity: displacement/time, vector (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Distance = 3 + 4 = 7 m (1); Displacement = sqrt(3 squared + 4 squared) (1); = 5 m (1). | 1+1+1 |
| 16 | Graph is a horizontal line (1); Area = 6 x 10 (1); = 60 m (1). | 1+1+1 |
| 17 | Distance = 50 km in 1 h (1); Average speed 50 km/h (1); Speedometer shows instantaneous speed (1). | 1+1+1 |
| | **OR:** v = at = 12 m/s (1); s = (1/2)at squared = 0.5 x 3 x 16 (1); = 24 m (1). | 1+1+1 |
| 18 | a = 20/10 = 2 m/s squared (1); s = (10 + 30)/2 x 10 (1); = 200 m (1). | 1+1+1 |
| | **OR:** v = u - gt gives 0 = 20 - 10t, t = 2 s (1); h = u squared / 2g = 400/20 (1); = 20 m (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | v = u + at from slope (1); s = area of trapezium = ut + (1/2)at squared (1); v squared = u squared + 2as by eliminating t (1); Meaning of symbols (1); Conditions: uniform acceleration (1). | 1+1+1+1+1 |
| | **OR:** Velocity along the tangent (1); Direction changes: accelerated (1); v = 2 x 3.14 x 0.5/2 (1); = 1.57 m/s (1); Speed constant (1). | 1+1+1+1+1 |
| 20 | Stage 1: a = 5 m/s squared (1); Stage 2: a = 0 (1); Stage 3: a = -6 m/s squared (1); Distance: 90 + 300 + 75 (1); = 465 m (1). | 1+1+1+1+1 |
| | **OR:** 72 km/h = 20 m/s; 54 km/h = 15 m/s (1); Relative speed 5 m/s (1); Distance to cover = 100 + 150 + 150 (1); = 400 m (1); Time = 80 s (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

- Using distance instead of displacement (or speed instead of velocity); for a round trip the displacement and average velocity are zero.
- Not converting km/h to m/s before using the equations of motion, or mixing km with seconds.
- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body $a$ is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.
- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.

- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still $g$.

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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# ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** uniform velocity | Velocity constant. |
| 2 | **(c)** zero | Instantaneously at rest. |
| 3 | **(a)** m/s | Metre per second. |
| 4 | **(b)** uniform acceleration | v increases uniformly. |
| 5 | **(b)** the direction keeps changing | Direction changes. |
| 6 | **(b)** m/s squared | Rate of change of velocity. |
| 7 | **(c)** 25 | 90 x 5/18. |
| 8 | **(a)** along the tangent | Tangent to the circle. |
| 9 | **(b)** 7 m | 3 + 4. |
| 10 | **(c)** 2 m/s squared | (20 - 10)/5. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Acceleration (1); Rise in velocity divided by time (1). | 1+1 |
| 12 | v = u + at (1); s = ut + (1/2)at squared and v squared = u squared + 2as (1). | 1+1 |
| 13 | No (1); Straight-line gap cannot exceed the path length (1). | 1+1 |
| 14 | Displacement (1); Area of a rectangle = v x t (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | t = 20/5 = 4 s (1); s = v squared / 2a = 400/10 (1); = 40 m (1). | 1+1+1 |
| 16 | a = (4 - 8)/2 = -2 m/s squared (1); Time to stop = 4/2 (1); = 2 s (1). | 1+1+1 |
| 17 | Scalar has magnitude only; vector has magnitude and direction (1); Scalars: distance, speed, time (1); Vectors: displacement, velocity, acceleration (1). | 1+1+1 |
| | **OR:** Average speed = 400/5 = 80 km/h (1); Net displacement = 0 (1); Average velocity = 0 (1). | 1+1+1 |
| 18 | Motion along a circle with constant speed (1); Examples: tip of a fan blade, Earth around the Sun (1); Velocity direction keeps changing (1). | 1+1+1 |
| | **OR:** Magnitude = sqrt(25 + 144) = 13 m (1); tan(theta) = 12/5 (1); About 67 degrees north of east (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Distance along the path (1); Displacement from the start (1); Speed = distance/time (1); Velocity = displacement/time (1); Acceleration due to changing direction (1). | 1+1+1+1+1 |
| | **OR:** Measure the distance (1); Time the student walking (1); Average speed = distance/time (1); Errors: reaction time (1); Repeat and average (1). | 1+1+1+1+1 |
| 20 | t up = 4 s (1); h = 1600/20 = 80 m (1); Total time 8 s (1); Velocity on return 40 m/s downward (1); Symmetry of motion (1). | 1+1+1+1+1 |
| | **OR:** h = 0.5 x 10 x 16 = 80 m (1); v = 40 m/s (1); Distance in 3 s = 45 m (1); Last second = 35 m (1); Using s = (1/2)gt squared (1). | 1+1+1+1+1 |

**Mistakes to avoid:**

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- Not converting km/h to m/s before using the equations of motion, or mixing km with seconds.
- Putting a positive sign for retardation in $v = u + at$ or $v^2 = u^2 + 2as$; for a slowing body $a$ is negative.
- Reading the slope of an x-t graph as acceleration and the slope of a v-t graph as velocity; slope of x-t is velocity, slope of v-t is acceleration.

- Calculating the distance from a v-t graph by adding signed areas or by taking only one segment; add the areas of all segments (triangles and rectangles) and take area below the axis as positive for distance.
- Thinking that acceleration is zero at the top of a vertical throw; only the velocity is zero, acceleration is still $g$.

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# ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 2 pi r / T | Circumference / time. |
| 2 | **(d)** 5 m | Pythagoras: sqrt(9 + 16). |
| 3 | **(b)** 50 km/h | 100/2. |
| 4 | **(c)** instantaneous speed | Speed at that instant. |
| 5 | **(a)** equal displacements in equal intervals of time in the same direction | Constant magnitude and direction. |
| 6 | **(c)** v = u + at | First equation. |
| 7 | **(d)** retardation (deceleration) | Velocity decreases. |
| 8 | **(d)** speed | Distance divided by time. |
| 9 | **(b)** s = ut + (1/2)at squared | Second equation. |
| 10 | **(a)** changes with time relative to a reference point | Motion is relative. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Uniform: equal distances in equal times (1); Non-uniform: unequal, e.g. a car in traffic (1). | 1+1 |
| 12 | 72 x 1000/3600 (1); = 20 m/s (1). | 1+1 |
| 13 | a = (20 - 10)/5 (1); = 2 m/s squared (1). | 1+1 |
| 14 | Total distance / total time (1); 5 m/s (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | h = 0.5 x 10 x 9 = 45 m (1); v = gt = 30 m/s (1); Starts from rest (1). | 1+1+1 |
| 16 | Total distance 50 m (1); Total time 5 s (1); Average speed 10 m/s (1). | 1+1+1 |
| 17 | Slope increases with time (1); Velocity is increasing (1); The body is accelerating (1). | 1+1+1 |
| | **OR:** A straight line sloping upward through the origin (1); Take two points and find rise/run (1); Slope = speed (1). | 1+1+1 |
| 18 | a = (v - u)/t (1); at = v - u (1); v = u + at (1). | 1+1+1 |
| | **OR:** Linear: a car on a straight road (1); Circular: a merry-go-round (1); Oscillatory: a pendulum (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Rest: horizontal s-t line (1); Uniform velocity: straight s-t line, horizontal v-t line (1); Uniform acceleration: straight rising v-t line (1); Retardation: falling v-t line (1); Slopes and areas (1). | 1+1+1+1+1 |
| | **OR:** Motion is relative (1); A passenger is at rest relative to the train (1); Moves relative to the platform (1); Trees seem to move backward (1); Describing motion needs a reference (1). | 1+1+1+1+1 |
| 20 | u = 5 m/s (1); v = 5 + 5 = 10 m/s (1); s1 = 75 m (1); Retardation = 0.5 m/s squared; s2 = 100 m (1); Total = 175 m (1). | 1+1+1+1+1 |
| | **OR:** Time going = 2 h; returning = 3 h (1); Total distance = 240 km (1); Total time 5 h (1); Average speed 48 km/h (1); Average speed is total distance over total time (1). | 1+1+1+1+1 |

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